Avishkar AV

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PROFESSIONAL SUMMARY

Fullstack Software Engineer with production experience architecting high-concurrency microservices, geospatial data engines, and scalable cloud platforms. Sole backend architect of bluedot.ngo (FastAPI, PostGIS, Cashfree 80G tax compliance) and core engineer of Passiton serving 16,000+ active users on GCP Cloud Run. Proven track record in database performance tuning, Redis in-memory caching, automated CI/CD pipelines, and regulatory data compliance (DPDP/GDPR).

WORK EXPERIENCE

Fullstack Developer, Third Eye Creative

Dec 2025 – Present | Bengaluru, India

- **BlueDot (Solo Backend & Platform Architect):** Architected the entire backend for bluedot.ngo as an asynchronous FastAPI REST API backed by PostgreSQL (Supabase) with SQLAlchemy 2.0 ORM, Pydantic schemas, and Alembic migrations.
- **PostGIS Geospatial Engine:** Stored restoration site boundary polygons (geometry(POLYGON, 4326)) and built automated PostgreSQL triggers with ST_Contains to increment cached tree counts without runtime aggregates.
- **Anti-Point-Farming Algorithm:** Engineered a spatial proximity rule using PostGIS ST_DWithin with a 5-meter threshold (within 5m counts as 15 XP verification; outside 5m creates 50 XP tree discovery).
- **Botanical AI Classification:** Integrated dual-engine AI vision (Vertex AI & PI@ntNet) analyzing up to 5 multi-angle images with 70% confidence threshold auto-publishing to the public Eco Garden Map.
- **Offline Donor Reconciler (Ghost User Engine):** Ingests offline donation CSVs, generates ghost accounts, and atomically merges historical records upon mobile OTP signup by phone number, retroactively awarding XP and badges.
- **FinTech & 80G Tax Compliance:** Integrated Cashfree Payment Gateway supporting webhook signature verification, delayed PAN collection UX flow, and automated 80G tax receipt PDF generation.
- **Legal & DPDP Act Compliance:** Engineered compliance architecture for India's DPDP Act and GDPR: document versioning, affirmative consent capture (logging IP, user-agent, timestamp), and self-service privacy export and erasure APIs.
- **Sub-50ms Redis Caching:** Built an in-memory caching layer for mobile leaderboards, computed via PostgreSQL window functions (RANK() OVER (ORDER BY total_impact_points DESC)).
- **Passiton (Marketplace for 16,000+ Users):** Shipped and maintained the production peer-to-peer recycling marketplace serving 16,000+ active users on GCP Cloud Run with autoscaling and zero-downtime updates.
- **Dual-Framework ASGI Architecture:** Architected a unified server hosting Flask at root (/) for the Admin Panel and FastAPI at /mobile for high-concurrency mobile APIs under a single ASGI container.
- **11-Stage State Machine:** Engineered an 11-stage item listing lifecycle state machine managing transitions across drafts, validation, requests, recipient selection, handoffs, and multi-quantity child cloning.
- **Database Performance Tuning:** Shrank production PostgreSQL database footprint **60% (2GB to 800MB)** and sped up complex join queries **40%** through zero-downtime poison-pill row cleanup and targeted indexing.
- **Latency & Cost Reduction:** Slashed feed response latency from **10s to <2s** with a Redis-cached feed algorithm, and eliminated 100% of third-party map costs by integrating OpenStreetMap.
- **AI Reliability Engineering:** Cut Vertex AI image-validation failure rates from **33% to 3%** through automated retry logic with exponential backoff in the marketplace ingestion pipeline.
- **Cloud Infrastructure Optimization:** Cut infrastructure spend **46%** for a legacy cash-collection tool by migrating from AWS EC2 to DigitalOcean droplets and containerizing the stack with Docker.
- **CI/CD Automation:** Built multi-tenant CI/CD pipelines using GitHub Actions and Docker Hub, accelerating build-to-delivery release cycles for 3 Play Store apps from hours to minutes.
- **NDB Government Portal:** Secured the NDB government portal for launch by implementing role-based access control (RBAC) and data masking policies.

Fullstack Developer Intern, EarthlyExotica

Jan 2025 – May 2025 | Bengaluru, India

- Built mobile-responsive, performant frontend interfaces in Flutter for baadige.com utilizing modular, reusable component architecture to accelerate UI development.
- Integrated core REST APIs to connect the Flutter mobile client to backend business logic, payment flows, and product catalog services.

TECHNICAL SKILLS

Languages & Backend: Python, SQL, JavaScript, TypeScript, Dart, Java, HTML5, CSS3, FastAPI, Flask, SQLAlchemy 2.0, Alembic, Pydantic, RESTful APIs, JWT Authentication, RBAC

Geospatial & Databases: PostGIS (ST_Contains, ST_DWithin, SRID 4326), PostgreSQL, Redis, Supabase, MySQL, MongoDB, Firebase, Spatial Indexing, OpenStreetMap

Cloud, DevOps & Infra: GCP (Cloud Run, Vertex AI, IAM), AWS (EC2), DigitalOcean, Docker, Docker Hub, GitHub Actions, Jenkins, Linux (Ubuntu/Debian), Vercel

Payment Gateways & APIs: Cashfree Payment Gateway (Webhooks, 80G Receipts), YouTube Data API v3, MSG91 (DLT SMS Templates, OTP), Cloudinary SDK

Mobile Development: Flutter, Dart, Android Studio, Cross-Platform UI Component Design

TECHNICAL PROJECTS

- **Autonomous YouTube Video Publishing Engine:** Architected a zero-cost, 100% serverless media pipeline that scripts, narrates, renders, and auto-publishes 1080p Shorts to YouTube daily via GitHub Actions CI/CD (5 AM IST cron) on headless Ubuntu runners with Xvfb virtual framebuffers. Automated YouTube Data API v3 publishing via headless OAuth 2.0 refresh-token handshake and resumable chunked video uploads (PUT protocol) consuming only **16% of daily free quota (1,600 / 10,000 units)**. Integrated Gemini Flash with 2-pass quality & visual-sync proofreading and Supabase DB deduplication, paired with edge neural speech synthesis (Kokoro-82M ONNX, Faster-Whisper int8) and deterministic 14 Mbps broadcast master H.264 video rendering via Node.js Chrome DevTools Protocol (CDP) and FFmpeg.
- **Smart Attendance System:** Real-time facial recognition pipeline on CCTV feeds that replaced manual logging with automated CSV attendance tracking using Python, OpenCV, and NumPy.
- **Weather Alert & SMS Notification System:** Automated service broadcasting localized rainfall alerts by SMS from real-time meteorological sensor data using Python, OpenWeatherMap API, and Twilio API.
- **Real-Time Sign Language Translation System:** Computer vision data-augmentation and noise-reduction pipeline translating sign language gestures into text in real time using Python, OpenCV, and ML.

EDUCATION

Bachelor of Engineering (B.E.) in Computer Science & Engineering — Dec 2021 – June 2025 | CGPA: 7.95 / 10.0

Visvesvaraya Technological University (VTU) | Bengaluru, India

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional
- Oracle OCI AI Foundations Associate
- Java Object-Oriented Programming (OOP) Certification